End-sem: worked solutions

10 Nov 2025 · 50 marks · 3 hours · CEPC12

Question 1

Q1 (a) 2 marks

> The surface tension of water in contact with air at 20 °C is 0.0725 N/m. The pressure inside a droplet of water is 0.02 N/cm² greater than outside. Find the diameter of the droplet.

For a droplet, $\Delta p = 4\sigma/d$; $\Delta p = 0.02$ N/cm² = 200 N/m²

$d = 4\sigma/\Delta p = 4 \times 0.0725 / 200 =$ **0.00145 m = 1.45 mm**

Q1 (b) 4 marks

> (i) A plate 0.025 mm from a fixed plate moves at 60 cm/s and needs a force of 2 N/m² to maintain this speed. Find the viscosity of the fluid between the plates.
>
> (ii) An open tank contains water to a depth of 2 m and above it oil of S = 0.9 to a depth of 1 m. Find the pressure intensity (i) at the interface and (ii) at the bottom.

(i)

$\tau = \mu\, du/dy \rightarrow 2 = \mu \times 0.6 / (0.025 \times 10^{-3})$

$\mu = 2 \times 0.025 \times 10^{-3} / 0.6 =$ **8.33×10^{-5} N·s/m² = 8.33×10^{-4} poise**

(ii)

At the interface: $p = \rho_{oil}\, g\, h_{oil} = 0.9 \times 9810 \times 1 =$ **8829 N/m² (8.83 kPa)**

At the bottom: $p = 8829 + 9810 \times 2 =$ **28,449 N/m² (28.4 kPa)** (gauge)

Q1 (c)

4 marks

> A differential manometer is connected at points A and B of two pipes as shown. Pipe A contains a liquid of S = 1.5, pipe B a liquid of S = 0.9. Pressures at A and B are 1 kgf/cm² and 1.80 kgf/cm². Find the difference in mercury level.

From the figure, take datum X–X at the **lower mercury surface** (right limb, B side). In the left limb the mercury stands h above X–X, with the S = 1.5 liquid above it rising (2 + 3) m to A. In the right limb the S = 0.9 liquid rises (h + 2) m from X–X to B.

$p_A = 1$ kgf/cm² $= 9.81 \times 10^4$ N/m²; $p_B = 1.8$ kgf/cm² $= 17.658 \times 10^4$ N/m².

Left limb at X–X: $p_A + 1500 \times 9.81 \times (2 + 3) + 13{,}600 \times 9.81 \times h = 98{,}100 + 73{,}575 + 133{,}416\ h$

Right limb at X–X: $p_B + 900 \times 9.81 \times (h + 2) = 176{,}580 + 17{,}658 + 8829\ h$

$133{,}416\ h - 8829\ h = 194{,}238 - 171{,}675 \rightarrow 124{,}587\ h = 22{,}563$

h = **0.181 m = 18.1 cm of mercury**

Question 2

Q2 (a)

2 marks

> Define (i) total pressure (ii) centre of pressure (iii) buoyancy.

- **Total pressure**: the resultant force exerted by a static fluid on a surface in contact with it, always acting **normal** to the surface: $F = \rho g A \bar{h}$.
- **Centre of pressure**: the point on the surface through which the **total pressure (resultant) acts**; for a plane surface it lies below the centroid: $h^* = \bar{h} + I_G \sin^2\theta / (A\bar{h})$.
- **Buoyancy**: the upward force exerted by a fluid on a body wholly or partly immersed in it, equal to the weight of fluid displaced (Archimedes).

Q2 (b)

4 marks

> (i) An oil of S = 0.9 and viscosity 0.06 poise flows through a pipe of 200 mm diameter at 60 l/s. Find the head lost due to friction in 500 m of pipe and the power required to maintain this flow.
>
> (ii) Find the total pressure and centre of pressure on an isosceles triangular plate of base 4 m and altitude 4 m immersed vertically in oil of S = 0.9, with the base at the free surface.

(i)

$V = Q/A = 0.06 / (\pi/4 \times 0.2^2) = 1.910$ m/s; $\mu = 0.06$ poise $= 0.006$ N·s/m²

$Re = \rho V d/\mu = 900 \times 1.910 \times 0.2 / 0.006 = 57{,}296 \rightarrow$ **turbulent**

$f = 0.079/Re^{0.25} = 0.079/15.47 = 0.0051$

$h_f = 4fLV^2/(2gd) = 4 \times 0.0051 \times 500 \times 1.910^2 / (2 \times 9.81 \times 0.2) =$ **9.49 m of oil**

Power $= \rho g Q h_f = 900 \times 9.81 \times 0.06 \times 9.49 =$ **5.03 kW**

(ii) Triangle with base at the surface, apex down. $A = \tfrac{1}{2} \times 4 \times 4 = 8$ m²; centroid at h/3 from the base, so $\bar{h} = 4/3 = 1.333$ **m**.

$F = \rho g A \bar{h} = 0.9 \times 9810 \times 8 \times 1.333 =$ **94.18 kN**

$I_G = bh^3/36 = 4 \times 4^3/36 = 7.111$ m⁴

$h^* = \bar{h} + I_G/(A\bar{h}) = 1.333 + 7.111/(8 \times 1.333) = 1.333 + 0.667 =$

2.0 m below the free surface

Q2 (c)

4 marks

> A solid cylinder of diameter 5.0 m and height 5.0 m has S = 0.7 and floats in water with its axis vertical. Find its metacentric height and state whether it is stable.

Depth of immersion: weight = buoyancy → $0.7 \times 5 = 1.0 \times h \rightarrow h =$ **3.5 m**

KB = 3.5/2 = 1.75 m; KG = 5/2 = 2.5 m → BG = 0.75 m

$BM = I/V = (\pi/64\, d^4) / (\pi/4\, d^2 \times 3.5) = d^2/(16 \times 3.5) = 25/56 = 0.446$ m

GM = BM − BG = 0.446 − 0.75 = **−0.304 m**

GM is **negative** (M lies below G), so the cylinder is in **unstable equilibrium** with its axis vertical; it will tip over and float on its side.

Question 3

Q3 (a)

2 marks

> Distinguish between (i) steady and unsteady flow, (ii) uniform and non-uniform flow, (iii) laminar and turbulent flow.

One type		The other type
Steady: velocity, pressure, density at a point do not change with time, $\partial V/\partial t = 0$ (constant discharge in a pipe)	vs	**Unsteady**: they change with time, $\partial V/\partial t \neq 0$ (flow while a tank empties)
Uniform: velocity does not change with distance along the flow at a given instant, $\partial V/\partial s = 0$ (constant-diameter pipe)	vs	**Non-uniform**: velocity changes along the flow (tapering pipe, river)
Laminar: particles move in smooth parallel layers, no mixing; viscous forces dominate; $Re < 2000$ in pipes	vs	**Turbulent**: particles move irregularly with eddies and mixing; inertia dominates; $Re > 4000$

Q3 (b)

4 marks

> State and derive Bernoulli's equation with assumptions.

Statement: for steady, ideal, incompressible flow along a streamline, the **total energy per unit weight** (pressure head + kinetic head + datum head) **is constant**: $p/\rho g + V^2/2g + z = \text{constant}$.

Derivation: take a small cylindrical element of length ds and area dA along a streamline inclined at θ. Forces along the streamline: net pressure force $-(\partial p/\partial s)\, ds\, dA$; weight component $-\rho g\, dA\, ds \cos\theta$, with $\cos\theta = dz/ds$. Mass × acceleration = $\rho\, dA\, ds \times V\, dV/ds$ (steady flow). So:

$-(\partial p/\partial s)\, ds\, dA - \rho g\, dA\, dz = \rho\, dA\, ds\, V\, dV/ds \Longrightarrow$ **$dp/\rho + g\, dz + V\, dV = 0$** (Euler)

Integrating with ρ constant: $p/\rho + gz + V^2/2 = \text{constant} \Longrightarrow$

$p/\rho g + V^2/2g + z = \text{constant}$

Assumptions: ideal (inviscid) fluid; steady flow; incompressible; irrotational; along a streamline; only gravity and pressure forces.

Q3 (c)

4 marks

> (i) The velocity potential of a 2-D potential flow is $\phi = 4x(3y - 4)$. Find the velocity at (2, 3) and the value of the stream function ψ at (2, 3).
>
> (ii) A 100 mm pipe has a nozzle (outlet 50 mm). The discharge is 20 l/s and the pressure at the nozzle base is 5.886 N/cm^2. Find the coefficient of discharge.

(i) $\phi = 12xy - 16x$. With $u = -\partial\phi/\partial x$ and $v = -\partial\phi/\partial y$:

$u = -(12y - 16)$, $v = -12x \rightarrow$ at (2, 3): $u = -20$, $v = -24$

Resultant velocity $V = \sqrt{(20^2 + 24^2)} =$ **31.24 units**

Stream function: $\partial\psi/\partial y = -u = 12y - 16 \rightarrow \psi = 6y^2 - 16y + f(x)$; $\partial\psi/\partial x = v = -12x \rightarrow f(x) = -6x^2$

$\psi = 6y^2 - 16y - 6x^2 \rightarrow$ at (2, 3): $\psi = 54 - 48 - 24 =$ **−18 units**

(With $u = +\partial\phi/\partial x$ the velocity components change sign but $V = 31.24$ and $\psi = -18$ are the same.)

(ii) $h = 58{,}860/9810 = 6$ m; $a_1 = 0.007854$ m^2, $a_2 = 0.001963$ m^2.

$Q_{th} = a_1 a_2 \sqrt{(2gh)}/\sqrt{(a_1^2 - a_2^2)} = 0.0220$ m^3/s

$C_d = 0.020 / 0.0220 =$ **0.909**

Question 4

Q4 (a)

2 marks

> Two pipes, each 2500 m long, of diameters 80 cm and 60 cm are connected in parallel. f = 0.006 for each. The total flow is 250 l/s. Find the flow in each pipe.

In parallel, the head loss is the same: $4fL\,V_1^2/(2g\,d_1) = 4fL\,V_2^2/(2g\,d_2) \rightarrow V_1 = V_2\sqrt{(d_1/d_2)} = 1.1547\,V_2$.

$Q = \pi/4\,(0.8^2\,V_1 + 0.6^2\,V_2) = \pi/4\,(0.64 \times 1.1547 + 0.36)\,V_2 = 0.8632\,V_2 = 0.25$

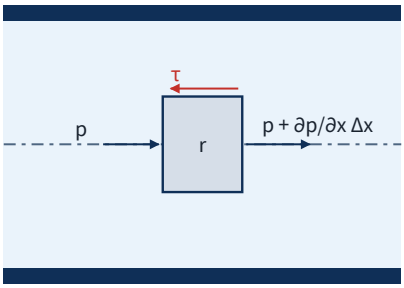
$V_2 = 0.2896$ m/s, $V_1 = 0.3344$ m/s

Q_1 (80 cm) $= \pi/4 \times 0.64 \times 0.3344 =$ **$0.168\ m^3/s$**; Q_2 (60 cm) $= \pi/4 \times 0.36 \times 0.2896 =$ **$0.082\ m^3/s$**

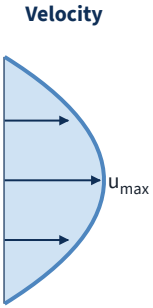
Q4 (b)

4 marks

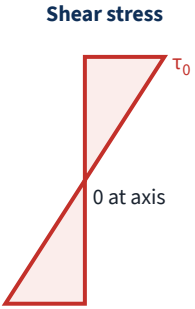
> Derive the Hagen–Poiseuille equation with neat sketches showing shear stress and velocity distribution across the pipe.



cylindrical element of radius r, length Δx



parabolic



linear

Shear stress. For a coaxial cylinder of radius r and length Δx in steady laminar flow, pressure force = shear force:

$$p\,\pi r^2 - (p + \partial p/\partial x\,\Delta x)\,\pi r^2 - \tau\,(2\pi r\,\Delta x) = 0 \implies \boldsymbol{\tau = (-\partial p/\partial x)\,r/2}$$

τ varies **linearly**: zero at the axis, maximum $\tau_0 = (-\partial p/\partial x)\, R/2$ at the wall.

Velocity. With $\tau = -\mu\, du/dr$ (u decreases as r increases):

$$-\mu\, du/dr = (-\partial p/\partial x)\, r/2 \Longrightarrow du = -(1/2\mu)(-\partial p/\partial x)\, r\, dr \Longrightarrow u = -(1/4\mu)(-\partial p/\partial x)\, r^2 + C$$

No slip, $u = 0$ at $r = R \to C = (1/4\mu)(-\partial p/\partial x)R^2$:

$$\boxed{\mathbf{u = (1/4\mu)(-\partial p/\partial x)(R^2 - r^2)}} \text{ (parabolic)}$$

u_{max} (at $r = 0$) $= (1/4\mu)(-\partial p/\partial x)R^2$. Discharge $Q = \int u\, 2\pi r\, dr = (\pi/8\mu)(-\partial p/\partial x)R^4$, so the **mean velocity $\bar{u} = Q/\pi R^2 = u_{max}/2$**.

Pressure drop. From $\bar{u} = (1/8\mu)(-\partial p/\partial x)R^2 = (1/32\mu)(-\partial p/\partial x)D^2$:

$$-\partial p/\partial x = 32\mu\bar{u}/D^2 \Longrightarrow \boxed{\mathbf{(p_1 - p_2)/\rho g = h_f = 32\, \mu\, \bar{u}\, L\, /\, (\rho\, g\, D^2)}}$$

Q4 (c)

4 marks

> A 300 mm pipe 1000 m long connects two reservoirs with a level difference of 15 m. Find the discharge. If an additional 300 mm pipe 600 m long is attached alongside the last 600 m of the existing pipe, find the increase in discharge. $f = 0.02$; neglect minor losses.

Taking $f = 0.02$ as the Darcy friction factor, $h_f = fLV^2/(2gd)$. $A = \pi/4 \times 0.3^2 = 0.0707\ m^2$.

Single pipe: $15 = 0.02 \times 1000 \times V_1^2 / (2 \times 9.81 \times 0.3) = 3.398\, V_1^2$

$V_1 = 2.101$ m/s $\to Q_1 = 0.0707 \times 2.101 =$ **$0.1485\ m^3/s$**

With the parallel pipe: first 400 m carries Q (velocity V); last 600 m has two pipes, each carrying $Q/2$ (velocity $V/2$):

$15 = (0.02 / (2 \times 9.81 \times 0.3))\, [400\, V^2 + 600\, (V/2)^2] = 0.003398 \times 550\, V^2 = 1.869\, V^2$

$V = 2.833$ m/s $\to Q_2 = 0.0707 \times 2.833 =$ **$0.2003\ m^3/s$**

Increase $= 0.2003 - 0.1485 =$ **$0.0518\ m^3/s$ (≈ 35% more)**

(If f is read as the coefficient in $h_f = 4fLV^2/2gd$, all discharges halve: $Q_1 = 0.0743$, $Q_2 = 0.1001$, increase 0.0259 m^3/s.)

Question 5

Q5 (a)

2 marks

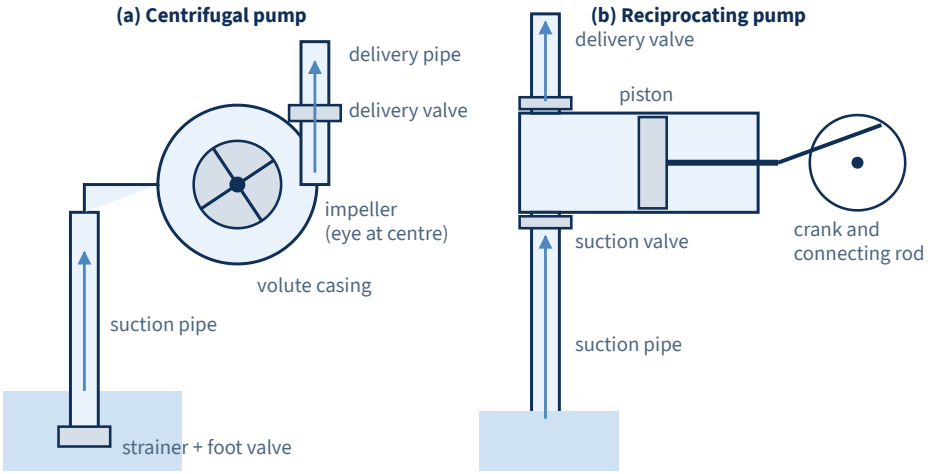
> Define (i) hydraulic efficiency, (ii) mechanical efficiency, (iii) overall efficiency of a turbine.

- **Hydraulic efficiency** η_h = power delivered **to the runner** ÷ power supplied at the inlet = $(\rho Q\, V_{w1}u_1 \pm V_{w2}u_2)/(\rho g Q H)$.
- **Mechanical efficiency** η_m = power at the **shaft** ÷ power delivered to the runner (accounts for bearing and friction losses).
- **Volumetric efficiency** η_v = water actually striking the runner ÷ water supplied.
- **Overall efficiency** η_o = shaft power ÷ water power at inlet = $P/(\rho g Q H)$ = $\boldsymbol{\eta_h \times \eta_m}$ ($\times\, \eta_v$).

Q5 (b)

4 marks

> With a neat sketch describe the working principle of a centrifugal pump and a reciprocating pump. Also explain the indicator diagram considering the effect of acceleration and velocity (friction).



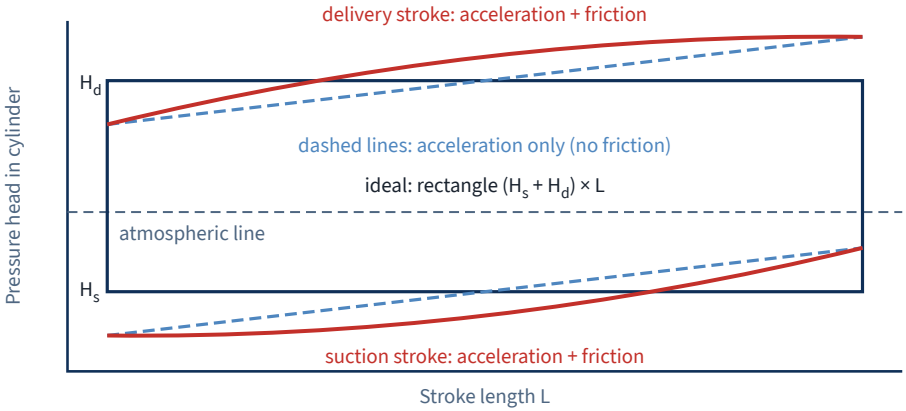
Centrifugal pump: a **rotodynamic** pump. Main parts: impeller with backward-curved vanes, volute casing, suction pipe with foot valve and strainer, delivery pipe with valve.

1. The pump is **primed** (casing and suction pipe filled with water, to remove air).
2. The rotating impeller produces a **forced vortex**; the centrifugal head ($\omega^2 r^2/2g$) raises the pressure towards the periphery.
3. The low pressure at the **eye** draws water up the suction pipe.
4. Water leaves the impeller with high velocity; the **volute casing** converts this kinetic energy into pressure, and water is delivered to the required height.

Reciprocating pump: a **positive-displacement** pump. Parts: cylinder with piston, crank and connecting rod, suction and delivery valves (non-return), suction and delivery pipes.

1. **Suction stroke**: the crank pulls the piston outward; the pressure in the cylinder falls below atmospheric; the suction valve opens and water enters.
2. **Delivery stroke**: the piston moves inward; pressure rises; the suction valve closes, the delivery valve opens and water is pushed into the delivery pipe.
3. Discharge per stroke = A × L (single-acting): $Q = ALN/60$. Suited to high heads, small discharges.

Indicator diagram (pressure head in the cylinder vs piston stroke):



- **Ideal**: a rectangle; suction at H_s below atmospheric and delivery at H_d above. Area $\propto$ work done per stroke = $\rho g A L(H_s + H_d)N/60$.
- **Effect of acceleration**: the water column in each pipe accelerates at the start of the stroke and decelerates at the end. The acceleration head $h_a = (l/g)(A/a)\omega^2 r \cos\theta$ is **maximum at the ends and zero at mid-stroke**. The suction and delivery lines become sloping straight lines (dashed): the cylinder pressure is $H_s + h_{as}$ below atmospheric at the start of suction and $H_s - h_{as}$ at the end; similarly for delivery. The areas added and removed are equal, so **work done is unchanged**.
- **Effect of friction (velocity)**: friction head $h_f = 4fl/(2gd)(A/a \times \omega r \sin\theta)^2$ is **zero at the ends and maximum at mid-stroke**, so parabolas are added (solid curves): extra suction at mid-suction and extra pressure at mid-delivery. Work done per stroke increases by ⅔ of the maximum friction heads: $W \propto H_s + H_d + \frac{2}{3}h_{fs} + \frac{2}{3}h_{fd}$.
- To avoid **separation (cavitation)**, the minimum pressure at the start of suction must stay above about 2.5 m of water (absolute); air vessels are used to reduce acceleration and friction heads.

Q5 (c)

4 marks

> A Francis turbine with overall efficiency 75% must produce 148.25 kW under a head of 7.62 m. Peripheral velocity = $0.26\sqrt{(2gH)}$; radial velocity of flow at inlet = $0.96\sqrt{(2gH)}$. The wheel runs at 150 rpm and the hydraulic losses are 22% of the available energy. Assuming radial discharge, find (i) the guide blade angle, (ii) the wheel vane angle at inlet, (iii) the wheel diameter at inlet, (iv) the width of the wheel at inlet.

$\sqrt{(2gH)} = \sqrt{(2 \times 9.81 \times 7.62)} = 12.227$ m/s

$u_1 = 0.26 \times 12.227 = 3.179$ m/s; $V_{f1} = 0.96 \times 12.227 = 11.738$ m/s

$\eta_h = 1 - 0.22 = 0.78$. With radial discharge ($V_{w2} = 0$): $\eta_h = V_{w1}u_1/(gH)$

$V_{w1} = 0.78 \times 9.81 \times 7.62 / 3.179 = 18.34$ m/s

(i) Guide blade angle $\tan\alpha = V_{f1}/V_{w1} = 11.738/18.34 = 0.640 \rightarrow$ **$\alpha = 32.6°$**

(ii) Vane angle at inlet $\tan\theta = V_{f1}/(V_{w1} - u_1) = 11.738/15.16 = 0.774 \rightarrow$ **$\theta = 37.7°$**

(iii) Diameter $u_1 = \pi D_1 N/60 \rightarrow D_1 = 60 \times 3.179/(\pi \times 150) =$ **0.405 m**

(iv) Width: $Q = P/(\eta_o \rho g H) = 148{,}250/(0.75 \times 9810 \times 7.62) = 2.644$ m^3/s

$Q = \pi D_1 B_1 V_{f1} \rightarrow B_1 = 2.644/(\pi \times 0.405 \times 11.738) =$ **0.177 m**